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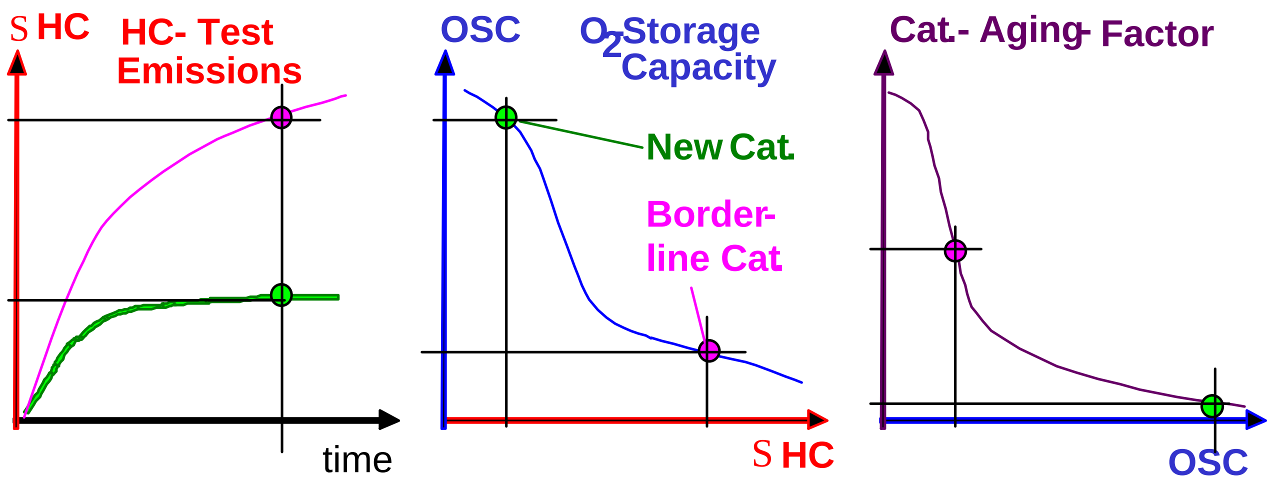
How to Measure

Oxygen Storage Capacity of a Catalyst

OSC Measurement Procedure

General Overview



Function of:

Light-Off
Driving
Cat. Aging
Emiss. Control

Function of:

Coating
Aging

Function of:

Lambda Sensor
Mambda Control
Monitoring Function

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Basics

- ➔ In a first step the catalyst is cleared from O_2 with a rich mixture (Lambda = 0.96).
- ➔ After that a rich mixture is applied, the catalyst will be filled with oxygen (lean mixture, Lambda =1.04), until the voltage of the downstream oxygen sensor jumps to lean.
- ➔ The amount of the needed oxygen ($0.23 \int ml^*\Delta\lambda$) supplies the oxygen storage capacity (OSC).
- ➔ A correlation to the emissions is only given, if there is a relation between stationary measured OSC and the conversion of HC and the light off of the catalyst

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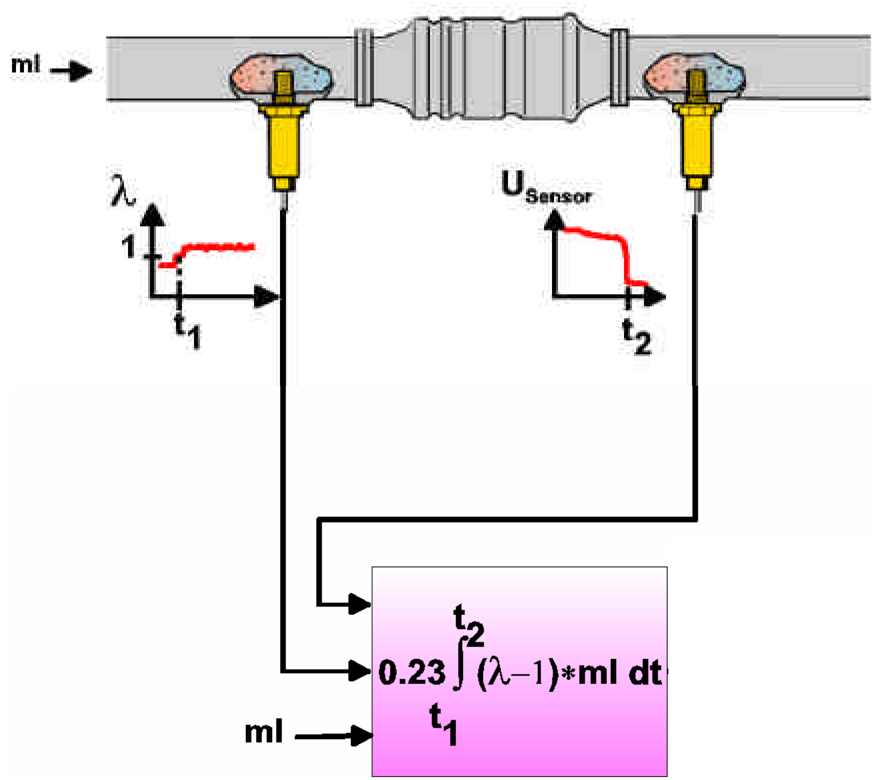
Measurement Principle



Measuring Conditions

- Measurements should be performed at ECE Cycle relevant working points (at least at 30km/h, 50 km/h and 70 km/h)
  Main focus should be on 70 km/h Measurements!
  Gears according to ECE Test Cycle!

- The system should be in steady state, that means engine speed, load and catalyst temperatures are stable .

- At each working point several measurements have to be taken to get a reliable average OSC for this working point

- After the catalyst has been cleared out of oxygen a jump from rich to lean is applied. From that point the time (t1) measurment starts and lasts until the downstream oxygen sensor voltage drops to lean (t2).

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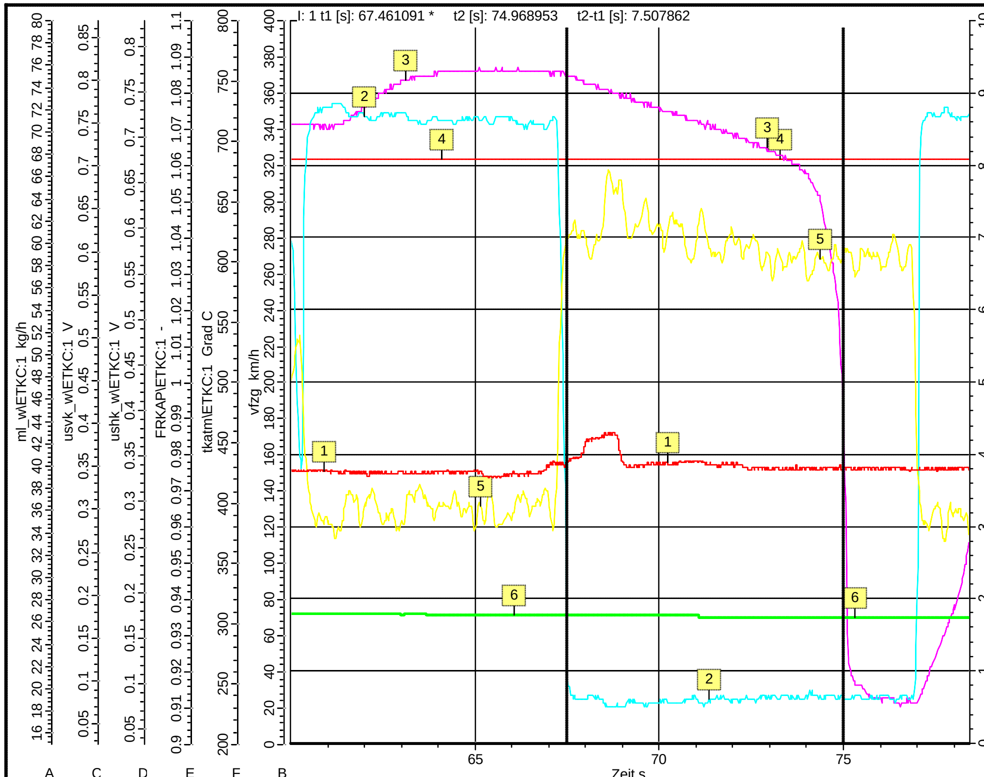
Example of Measurement with new Catalyst



	Lini	Name	Cur	Cur	Curs	Ei	Pro-	Star
		ml_w	39.	39.7	-0.2	kg	6.5	15
		usvk_w	0.3	0.08	-	V	0.08	0.02
		ushk_w	0.7	0.43	-	V	0.07	0.03
		tikatm	685	685	0	Gr	60	200
		LA4:1_...	1.0	1.03	-		0.02	0.9
		vfzg	71.	70	-1.25	k	40	0

	Datum: Zeit: Anw ender: Einheit:	Objekt: Projekt: Messdatei:

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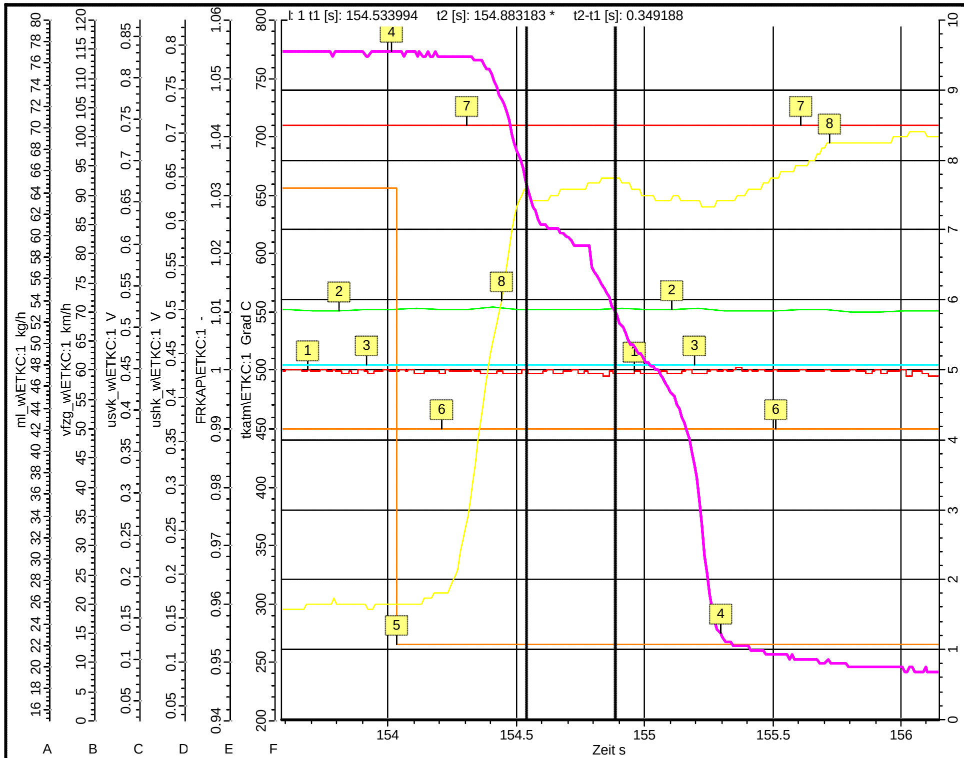
Example of Measurement with aged Catalyst



	Lini	Name	Cur	Cur	Cur	Ei	Pro-	Star
		ml_w	47.	47.	-0.4	kg	6.5	15
		vfzg_w	70.	70.	0.08	km	12	0
		usvk_w	0.4	0.4	0	V	0.08	0.02
		ushk_w	0.6	0.4	-	V	0.07	0.03
		FRKAP	0.9	0.9	0	-	0.01	0.94
		tkatm	450	450	0	Gr	60	200
		tikatm	710	710	0	Gr	60	200
		LA4:1_L...	1.0	1.0	0.00		0.01	0.94

	Datum: Zeit: Anw ender: Einheit:	Objekt: Projekt: Messdatei:

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Calculation of OSC from Measurements
(Refer to Example of new Catalyst)

Input:

- ➔ Airmass: 39 kg/h
- ➔ Average Lambda: 1.03
- ➔ t2-t1: 74.97s – 67.46s = 7.51s

Output: Calculated OSC

➔ **OSC** = ((average Lambda-1) * (t2-t1) * airmass * 0.23 *1000) / 3.6
= **561.37 mg**

Calculation of OSC from Measurements
(Refer to Example of aged Catalyst)

Input:

- ➔ Airmass: 47 kg/h
- ➔ Average Lambda: 1.03
- ➔ t2-t1: 154.88s – 154.53s = 0.35s

Output: Calculated OSC

➔ **OSC** = ((average Lambda-1) * (t2-t1) * airmass * 0.23 *1000) / 3.6
= **31.53 mg**

Example for Measurements Results of a Borderline Catalyst with this Measurement Procedure

OSC Measurement	
Messung:	OSC [mg]
Aged-CAT-70km.dat	**30.84**
Aged-CAT-100km.dat	**27.86**
Aged-CAT-50km.dat	**25.89**
Aged-CAT-30km.dat	**41.23**